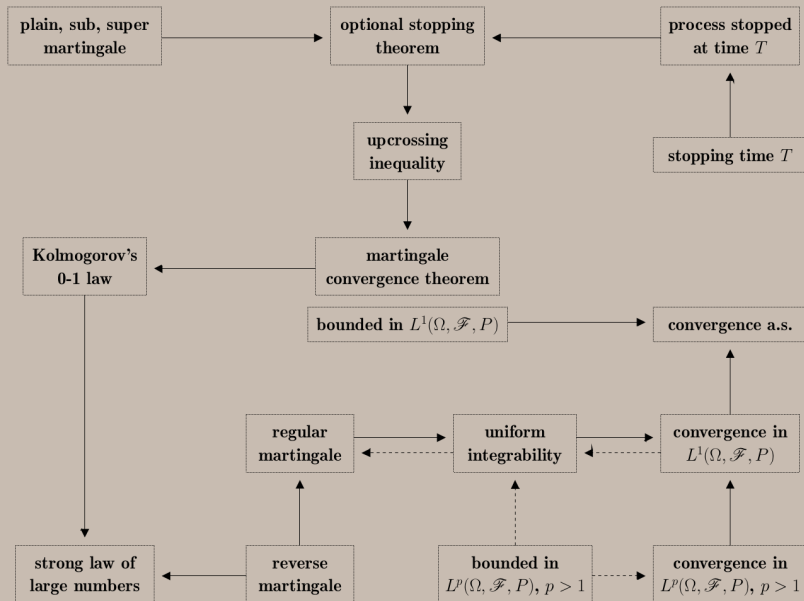
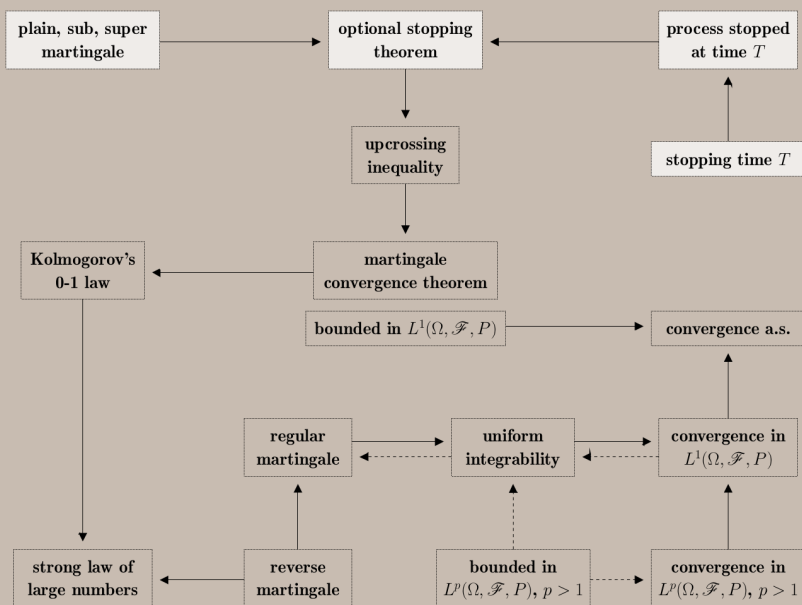




Paul Lévy



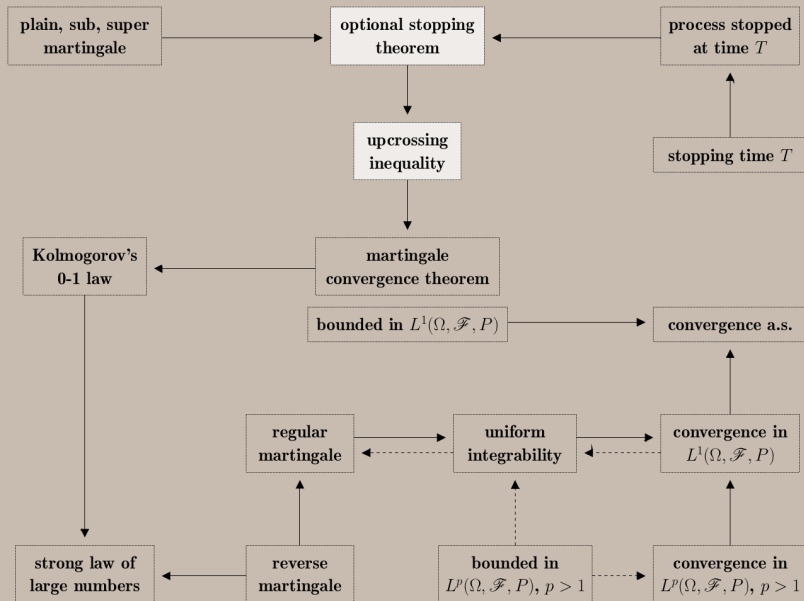
Joseph Doob



Paul Lévy



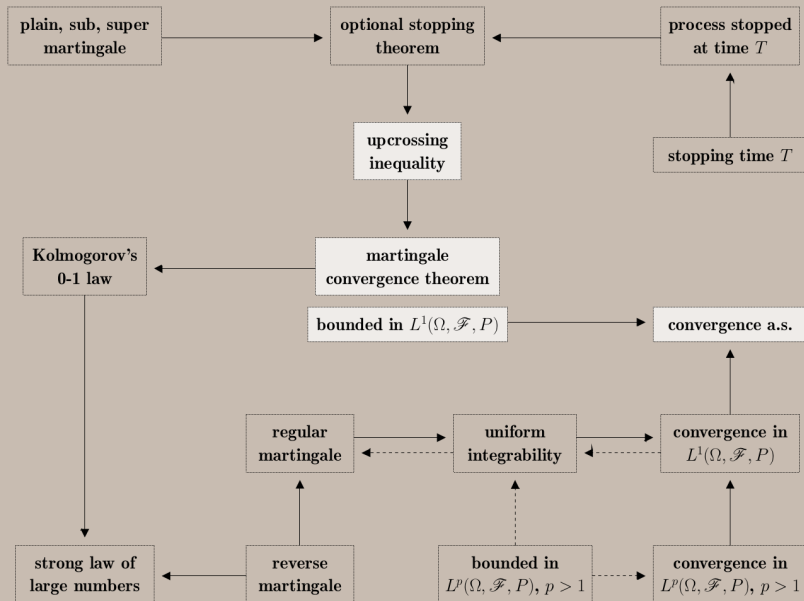
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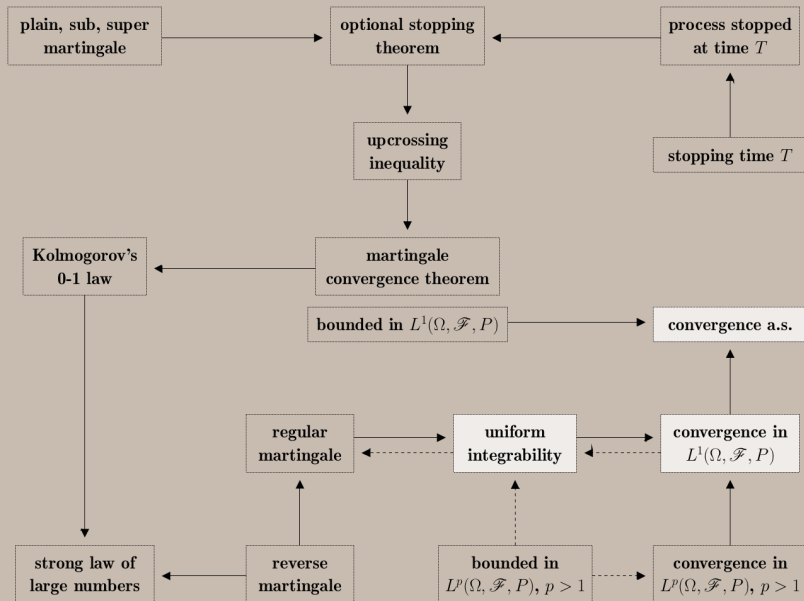
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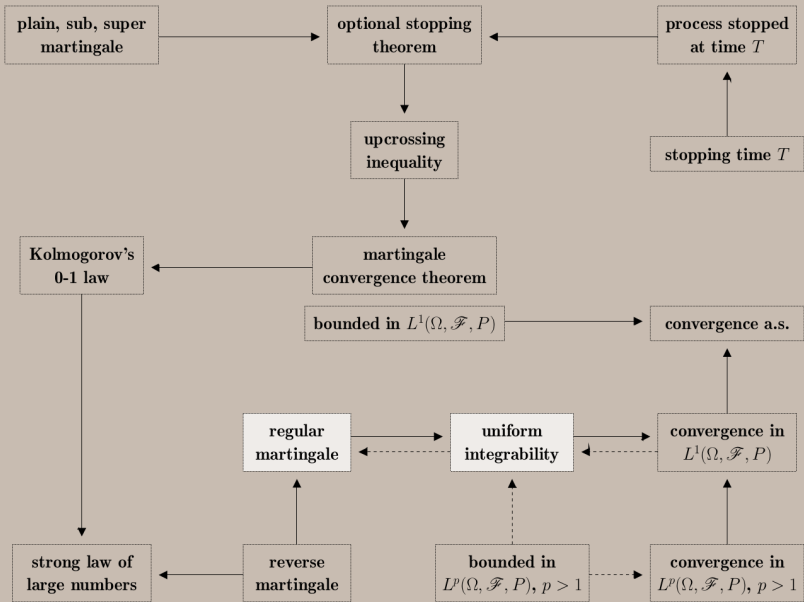
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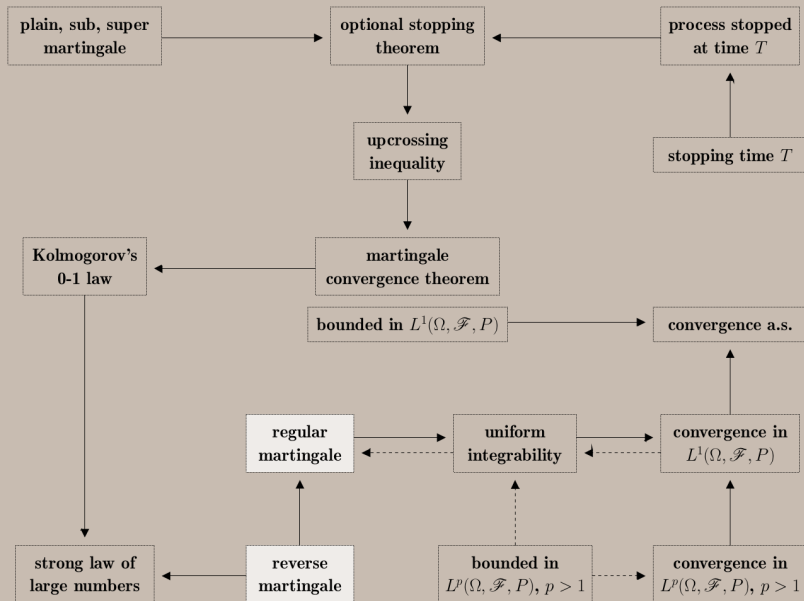
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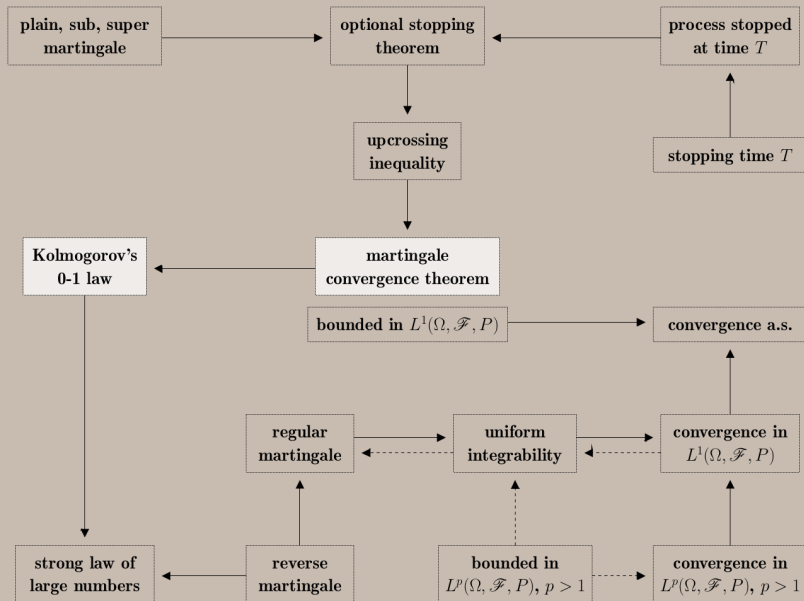
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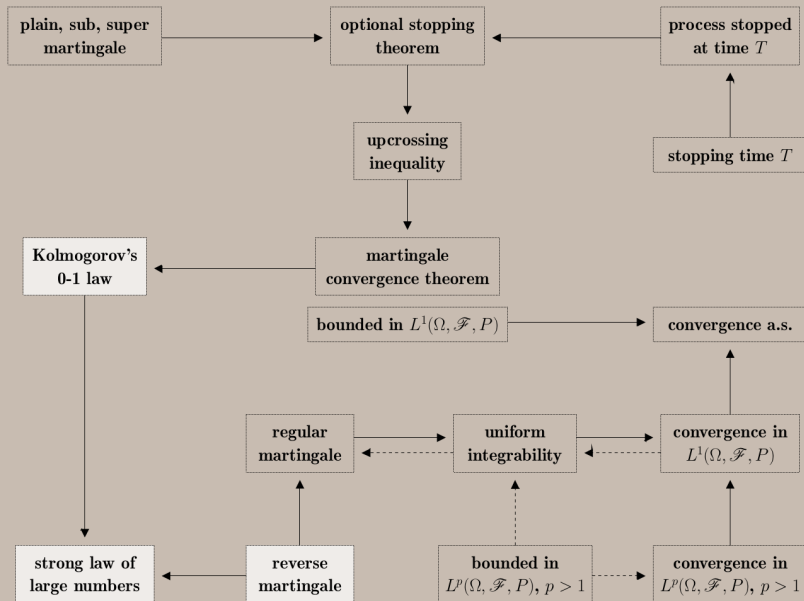
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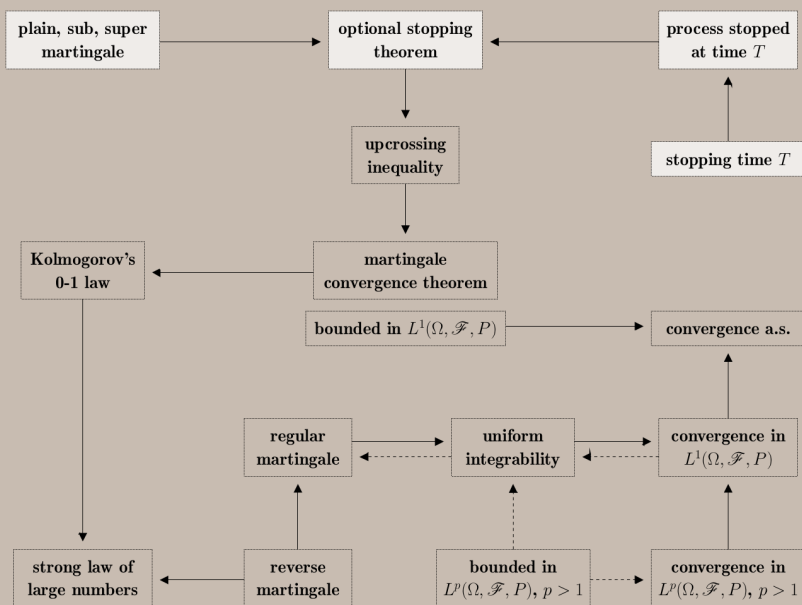
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$$1) X_n \in L^1(\Omega, \mathcal{F}, P) \text{ is adapted to } \mathcal{G}_n \quad 2) E(X_{n+1} | \mathcal{G}_n) \leq X_n.$$

Think of X_n as the fortune of a gambler playing an **unfavorable game**.



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$$T : \Omega \longrightarrow \bar{\mathbb{N}} = \mathbb{N} \cup \{\infty\} \quad \text{such that} \quad \{T = n\} \in \mathcal{G}_n \text{ for all } n \geq 0.$$

Interpretation: Decision to stop at time T cannot be based on the future.



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The process $X_n^T = X_{n \wedge T}$ **stopped** at time T is again a **supermartingale**:

$$\begin{aligned} E(X_{n+1}^T - X_n^T | \mathcal{G}_n) &= E((X_{n+1} - X_n) \mathbf{1}\{T > n\} | \mathcal{G}_n) \\ &= \mathbf{1}\{T > n\} E(X_{n+1} - X_n | \mathcal{G}_n) \leq 0. \end{aligned}$$



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$$\begin{aligned} E|Y| - E|X_0| &\stackrel{\text{MCT}}{=} \sum_k E(|X_{k+1} - X_k| \mathbf{1}\{T > k\}) \\ &\leq c \sum_k P(T > k) = c \sum_k \sum_{j > k} P(T = j) \\ &\stackrel{\text{FUB}}{=} c \sum_j \sum_{k < j} P(T = j) = c \sum_j j P(T = j) = c E(T) < \infty. \end{aligned}$$



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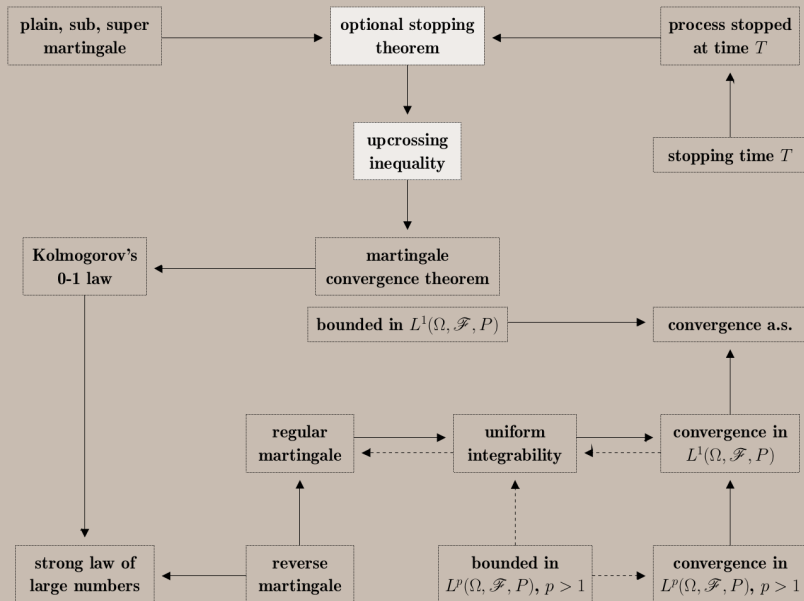
We have $E(T) = 2$ but the increments are not bounded.



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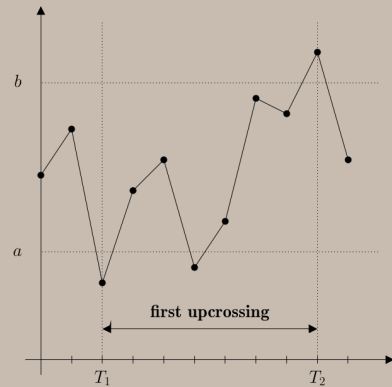


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Let $a < b$ and $u(a, b) = \#$ **upcrossings** of (a, b) .



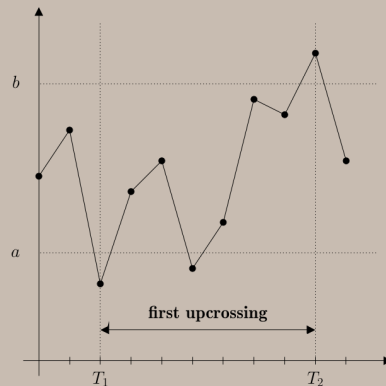
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Letting $T_0 = 0$, and the **stopping times**

$$T_{2k+1} = \inf \{n \geq T_{2k} : X_n \leq a\},$$

$$T_{2k+2} = \inf \{n \geq T_{2k+1} : X_n \geq b\},$$

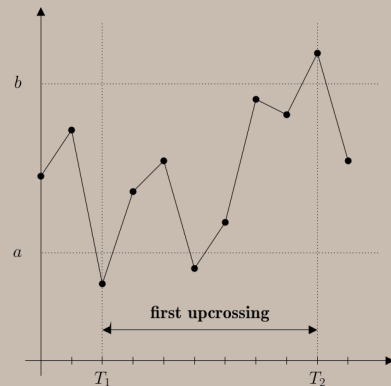
we have $u(a, b) = \sup \{k : T_{2k} < \infty\}$.



Let $u_N(a, b) = \#$ **upcrossings** of (a, b) by time N .

Upcrossing inequality. For all $k \geq 0$,

$$P(u_N(a, b) > k) \leq E((X_N - a)^- \mathbf{1}\{u_N(a, b) = k\}) / (b - a).$$

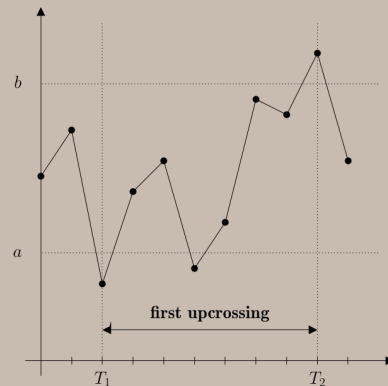


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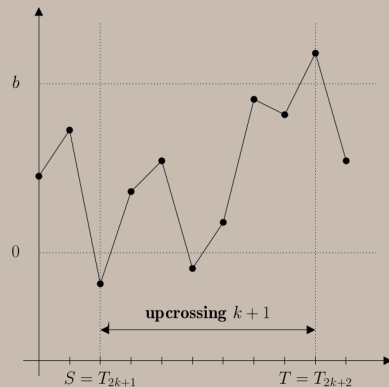
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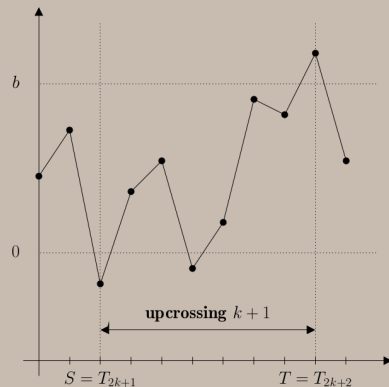
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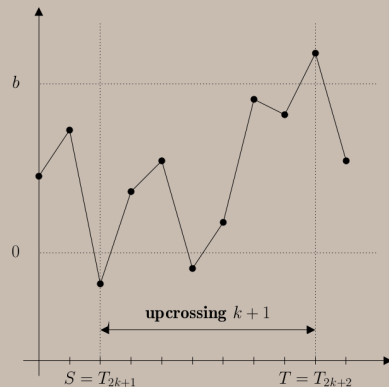
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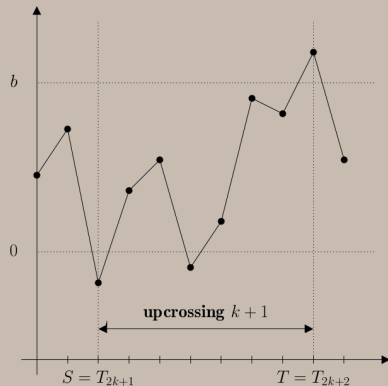
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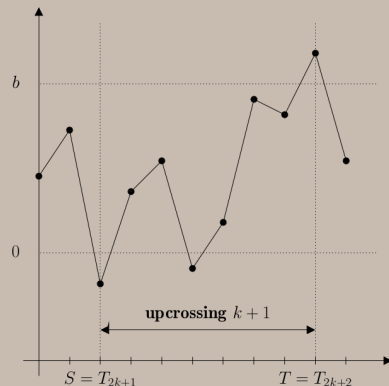
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Optional stopping theorem $\implies$

$$\begin{aligned} E(X_{T \wedge N}^+ \mathbf{1}\{S \leq N\}) - E(X_{T \wedge N}^- \mathbf{1}\{S \leq N\}) \\ \stackrel{\text{LIN}}{=} E((X_{T \wedge N}^+ - X_{T \wedge N}^-) \mathbf{1}\{S \leq N\}) = E(X_{T \wedge N} \mathbf{1}\{S \leq N\}) \stackrel{\text{OST}}{\leq} E(X_S \mathbf{1}\{S \leq N\}) \leq 0. \end{aligned}$$



Let $u_N(a, b) = \#$ **upcrossings** of (a, b) by time N .

Upcrossing inequality. For all $k \geq 0$,

$$P(u_N(a, b) > k) \leq E((X_N - a)^- \mathbf{1}\{u_N(a, b) = k\}) / (b - a).$$

Proof. $X_n - a$ supermartingale so assume $a = 0$.

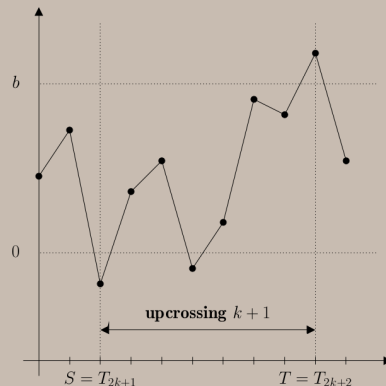
$$\{u_N(0, b) > k\} = \{T \leq N\} \subset \{S \leq N\} \cap \{X_{T \wedge N} \geq b\}.$$

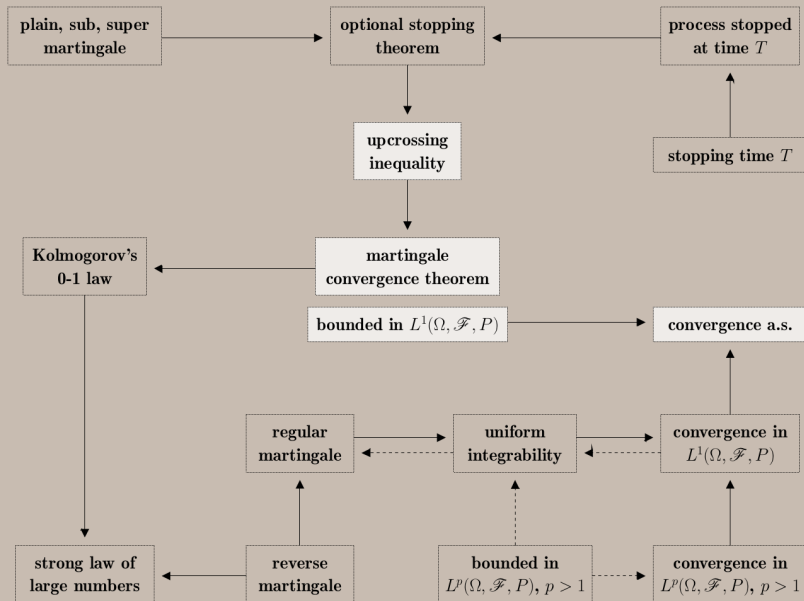
$$\begin{aligned} \implies b P(u_N(0, b) > k) &\stackrel{\text{LIN}}{=} E(b \mathbf{1}\{u_N(0, b) > k\}) \\ &\leq E(X_{T \wedge N} \mathbf{1}\{u_N(0, b) > k\}) \\ &\leq E(X_{T \wedge N}^+ \mathbf{1}\{S \leq N\}). \end{aligned}$$

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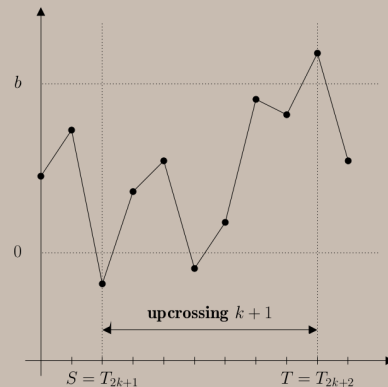
Paul Lévy



Joseph Doob

Martingale convergence theorem. For a supermartingale such that $\sup_n E|X_n| = M < \infty$,

$$X_n \xrightarrow{a.s.} X_\infty \quad \text{where} \quad X_\infty \in L^1(\Omega, \mathcal{F}, P).$$

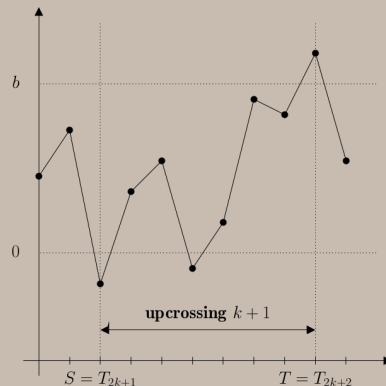


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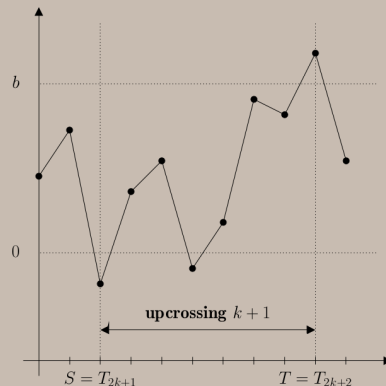
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Note: $u_N(a, b) \uparrow u(a, b)$

$$\implies E(u(a, b)) \stackrel{\text{MCT}}{=} \lim_{N \rightarrow \infty} E(u_N(a, b)) < \infty$$

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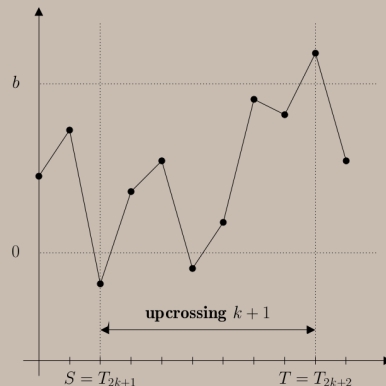
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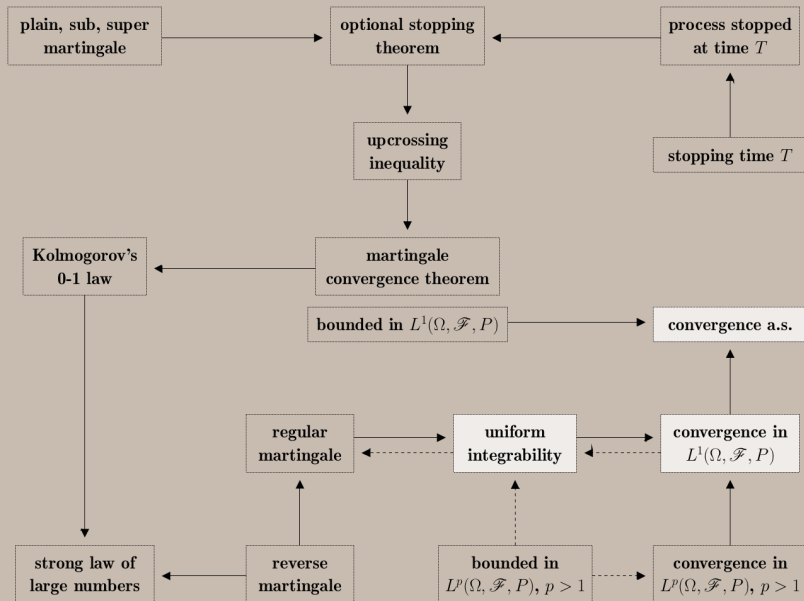
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Because $\bar{\mathbb{Q}} = \mathbb{R}$ and $\mathbb{Q}$ is countable,

$$\begin{aligned} P(\liminf_n X_n < \limsup_n X_n) &= P(\liminf_n X_n < a < b < \limsup_n X_n \text{ for some } a, b \in \mathbb{R}) \\ &= P(\liminf_n X_n < a < b < \limsup_n X_n \text{ for some } a, b \in \mathbb{Q}) \\ &= P(u(a, b) = \infty \text{ for some } a, b \in \mathbb{Q}) = 0. \end{aligned}$$





Paul Lévy



Joseph Doob

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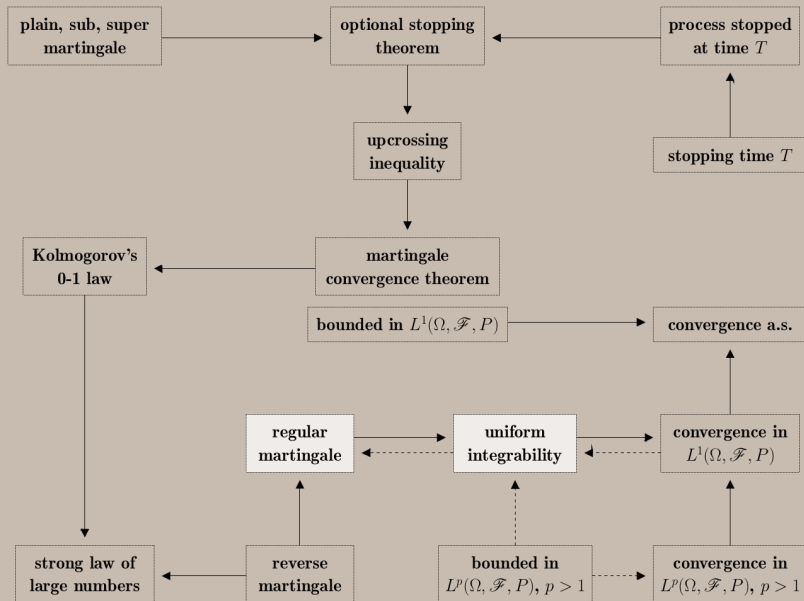
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Paul Lévy

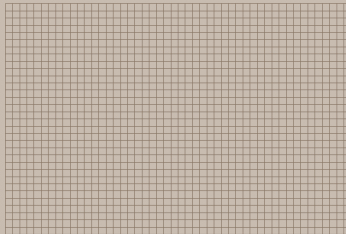


Joseph Doob

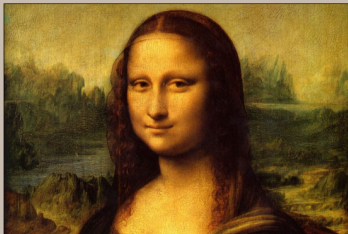
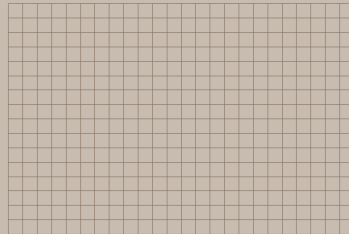
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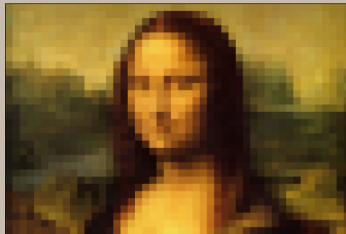
Sub- σ -algebra $\mathcal{G}_{n+1}$



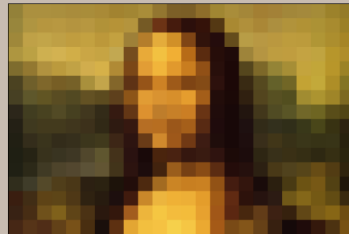
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Random variable X



$X_{n+1} = E(X | \mathcal{G}_{n+1})$



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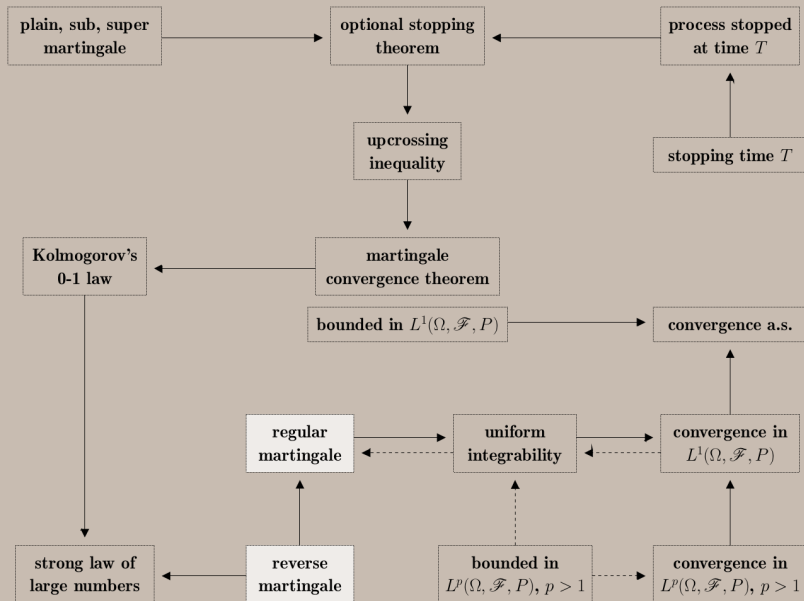
Let $M \geq E|X|/\delta$. **Markov's inequality** $\implies$

$$\begin{aligned} P(E(|X| | \mathcal{G}_n) > M) &\stackrel{\text{MI}}{\leq} E(E(|X| | \mathcal{G}_n))/M = E|X|/M \leq \delta \\ &\implies E(|X| \mathbf{1}\{E(|X| | \mathcal{G}_n) > M\}) \leq \epsilon. \end{aligned}$$

Jensen's inequality $\implies |X_n| = |E(X | \mathcal{G}_n)| \stackrel{\text{JI}}{\leq} E(|X| | \mathcal{G}_n)$

$$\begin{aligned} \implies E(|X_n| \mathbf{1}\{|X_n| > M\}) &\leq E(E(|X| | \mathcal{G}_n) \mathbf{1}\{E(|X| | \mathcal{G}_n) > M\}) \\ &= E(|X| \mathbf{1}\{E(|X| | \mathcal{G}_n) > M\}) \leq \epsilon \end{aligned}$$

$\implies X_n$ uniformly integrable.



Paul Lévy



Joseph Doob

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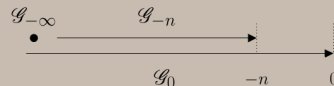
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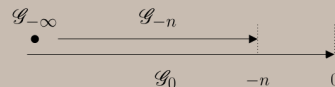
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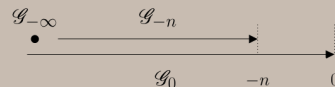
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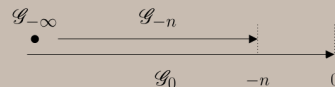
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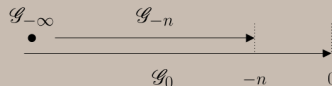
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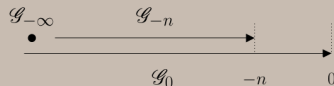
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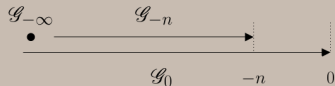
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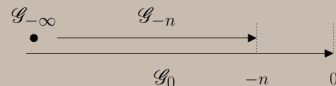
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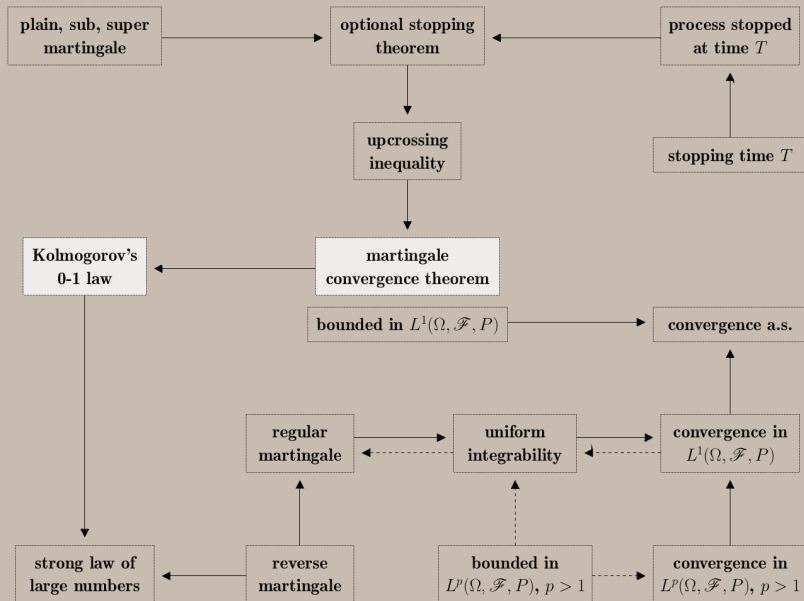
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Paul Lévy



Joseph Doob

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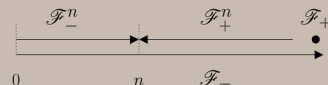
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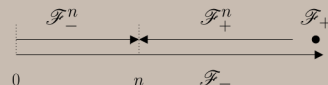
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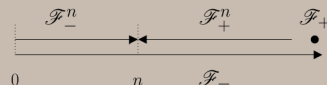
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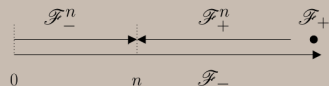
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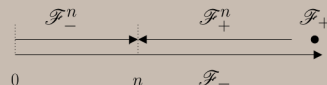
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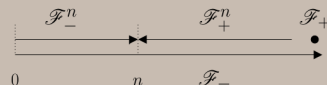
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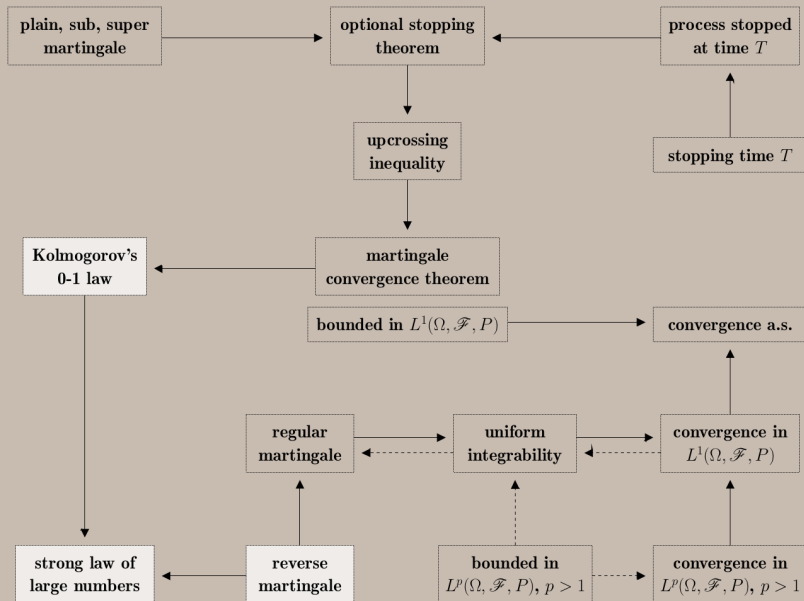
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Joseph Doob

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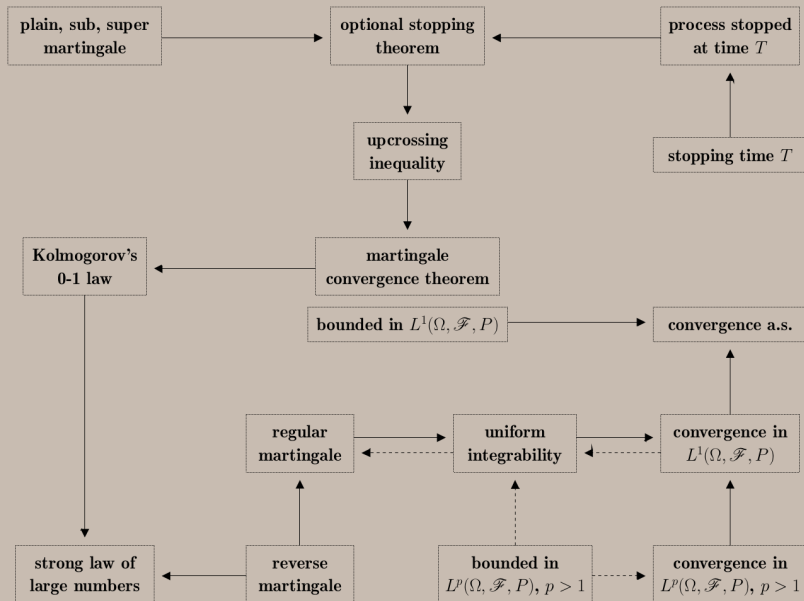
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Kolmogorov's zero-one law $\implies Y_{-\infty} \stackrel{\text{K01}}{=} E(Y_{-\infty}) = E(E(Y_{-1} | \mathcal{G}_{-\infty})) = E(Y_{-1}) = E(X_1) = \mu.$



Paul Lévy



Joseph Doob